

ASCII MASTER FLOW – PANEL 1/12: Temperate grassland world map

EURASIA -> steppes: Ukraine-Kazakhstan belt, continental interior

NORTH AMERICA -> prairies: Great Plains, USA and Canada

SOUTH AMERICA -> pampas: Argentina and Uruguay, maritime fringe

SOUTH AFRICA -> veld: elevated interior plateau

AUSTRALIA -> downs: eastern interior grasslands

MUST REMEMBER: Temperate continental steppe Bs climates occupy interiors and rain shadows between deserts and humid forests, with low variable precipitation, large annual range, short grasses and chernozem-linked grain/pastoral economies.

ASCII MASTER FLOW – PANEL 2/12: Continentality mechanism

DISTANCE FROM SEA -> weak maritime moderation

SUMMER -> strong heating of continental mass -> hot

WINTER -> strong radiative cooling -> cold

RESULT -> large annual temperature range, especially in Eurasia

ASCII MASTER FLOW – PANEL 3/12: Rainfall-moisture profile

TOTAL ANNUAL -> about 25-75 cm, low to moderate

SEASONAL MAXIMUM -> late spring and early summer

DRY MARGIN -> shorter grass, steppe-desert transition

WET MARGIN -> taller grass, steppe-forest boundary

ASCI MASTER FLOW – PANEL 4/12: Hemisphere asymmetry

NORTHERN -> steppes and prairies: large continental mass, extreme range

SOUTHERN -> pampas, veld, downs: narrower landmass, maritime moderation

CAUSE -> continental mass determines winter severity

EXAM TRAP -> do not treat all five as climatically identical

ASCII MASTER FLOW – PANEL 5/12: Chernozem-soil chain

GRASSLAND COVER -> deep root network decays in place

LOW LEACHING -> base-rich conditions preserved

HUMUS ACCUMULATES -> dark, deep, fertile soil profile

RESULT -> chernozem or black earth, among world's most fertile

ASCII MASTER FLOW – PANEL 6/12: Granary conversion timeline

NATURAL STATE -> pastoral grassland, low-density use

RAIL + OCEAN FREIGHT -> access to distant industrial markets

MECHANISATION + HYV -> high output per worker on level plains

RESULT -> world's wheat granaries: Ukraine, prairies, pampas

ASCII MASTER FLOW – PANEL 7/12: Anti-determinism firewall

PHYSICAL BASE -> fertile soil + level relief + suitable climate

CONVERSION AGENTS -> rail, freight, mechanisation, market demand

WITHOUT ACCESS -> same soil, no granary for centuries

CONCLUSION -> endowment is necessary, access is sufficient

ASCII MASTER FLOW – PANEL 8/12: Ecological-cost chain

SOD REMOVAL -> bare fine soil exposed to wind

DROUGHT YEAR -> vegetation fails, deflation peaks

ORGANIC DECLINE -> reduced humus, compaction risk

AQUIFER DRAWDOWN -> irrigation in semi-arid belt depletes groundwater

ASCII MASTER FLOW – PANEL 9/12: India wheat granary map

PUNJAB-HARYANA-WESTERN UP -> Indo-Gangetic alluvial plain

SOIL -> alluvial, not chernozem (fundamental difference)

SEASON -> Rabi: sown Oct-Dec, harvested Mar-Apr

ENABLERS -> HYV seeds + irrigation + fertiliser + MSP procurement

ASCII MASTER FLOW – PANEL 10/12: Green Revolution package

HYV SEMI-DWARF WHEAT -> raised yield potential

CANAL + TUBE-WELL IRRIGATION -> reduced rainfall dependence

CHEMICAL FERTILISERS -> supported intensive cultivation

MSP/PROCUREMENT -> assured floor price and central pool stocks

CLOSE DISTINCTION: Steppe differs from tropical savanna and humid prairie, chernozem is not universal across all steppes, and continentality concerns land-dominated thermal range rather than simply distance from a national coastline.

ASCII MASTER FLOW – PANEL 11/12: Savanna-steppe comparison

LATITUDE -> tropical savanna versus temperate steppe/prairie

RAINFALL -> wet-and-dry seasonal versus low-total spring maximum

SOIL -> leached laterising versus base-rich chernozem/prairie

ECONOMY -> pastoral-subsistence versus mechanised-commercial grain

EVIDENCE LIMIT: Grain productivity depends on soil, snowmelt, technology and markets as well as climate; India's wheat belt is a functional comparison with monsoonal irrigation and subtropical seasonality, not a climatic identity.

ASCII MASTER FLOW – PANEL 12/12: Steppe answer spine

DEFINE -> continental interior grassland with large annual range

LOCATE -> five regional names on a world map

EXPLAIN -> moisture deficit, chernozem, granary conversion

QUALIFY -> anti-determinism: access converted endowment into resource